

1 Knot 4_1 and its Arithmetic Interpretation

The figure-eight knot, denoted 4_1 , is a fundamental example in knot theory. Its group presentation can be given with:

- Generators: a, b
- Relator: $abbbaBAAB = 1$ (where $A = a^{-1}, B = b^{-1}$)
- Alexander polynomial: $\Delta(t) = t^2 - 3t + 1$

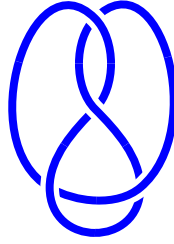


Figure 1: The Figure-Eight Knot (4_1)

In the framework of arithmetic expression geometry, we can interpret the generators and their inverses as operators acting within an arithmetic expression space H . We establish the following mapping, linking the knot group structure to arithmetic operations involving an indeterminate variable t (associated with the Alexander polynomial):

- $a \mapsto \otimes_t$ (multiplication by t)
- $b \mapsto \oplus_1$ (addition of 1)
- $A = a^{-1} \mapsto \oslash_t$ (division by t , i.e., multiplication by t^{-1})
- $B = b^{-1} \mapsto \ominus_1$ (subtraction of 1)

Under this interpretation, the relator $abbbaBAAB$ corresponds to a closed loop path (a specific threadlike expression) in the space H .

Figure 2: Schematic representation of the relator path in the arithmetic expression space H

Applying the sequence of operators corresponding to the relator $abbbaBAAB$ to a real number x (acting from right to left, consistent with function composition)

tion) yields:

$$\begin{aligned}
\text{Result} &= a(b(b(b(a(B(A(A(B(x)))))))))) \\
&= a(b(b(b(a(B(A(A(x-1)))))))) \\
&= a(b(b(b(a(B(A((x-1)t^{-1}))))))) \\
&= a(b(b(b(a(B((x-1)t^{-2})))))) \\
&= a(b(b(b(a((x-1)t^{-2}-1)))))) \\
&= a(b(b(b(((x-1)t^{-2}-1)t)))) \\
&= a(b(b(b((x-1)t^{-1}-t)))) \\
&= a(b(b((x-1)t^{-1}-t+1))) \\
&= a(b((x-1)t^{-1}-t+2)) \\
&= a((x-1)t^{-1}-t+3) \\
&= ((x-1)t^{-1}-t+3)t \\
&= (x-1)-t^2+3t \\
&= x-(t^2-3t+1)
\end{aligned}$$

Since the relator $abbbbaBAAB$ is equal to the identity element in the knot group, the corresponding path acting on x must return x . Therefore, we must have:

$$x - (t^2 - 3t + 1) = x$$

This forces the condition $t^2 - 3t + 1 = 0$, which is precisely the Alexander polynomial $\Delta(t)$ of the knot 4_1 set to zero. Solving this equation yields the roots $t = \frac{3 \pm \sqrt{5}}{2}$, which are related to the square of the golden ratio ϕ (i.e., $t = \phi^2$ or $t = \phi^{-2}$).

1.1 Computational Verification and Generalization

This observed connection between the knot relator, interpreted as a sequence of arithmetic operations, and the Alexander polynomial equation $\Delta(t) = 0$ is not unique to the knot 4_1 . Computational checks were performed using SageMath with the **snappy** library for knots with up to 11 crossings.

The procedure involved mapping generators (typically denoted a, b) of the knot group presentation to operators $\otimes_t$ and $\oplus_1$ (and their inverses $A \mapsto \otimes_t, B \mapsto \oplus_1$), applying the operator sequence corresponding to a group relator to a variable x , and verifying if the condition for the path to be closed ($\text{Result} = x$) precisely yields the Alexander polynomial equation $\Delta(t) = 0$ (up to factors of $\pm t^k$).

This direct correspondence was verified for the following specific knots within the checked range:

- 3 crossings: None
- 4 crossings: 4_1

- 5 crossings: None
- 6 crossings: $6_2, 6_3$
- 7 crossings: $7_6, 7_7$
- 8 crossings: $8_2, 8_9, 8_{10}, 8_{12}$
- 9 crossings: $9_{11}, 9_{17}, 9_{26}, 9_{27}, 9_{42}, 9_{44}$
- 10 crossings: $10_2, 10_{17}, 10_{41}, 10_{43}, 10_{44}, 10_{45}, 10_{47}, 10_{127}, 10_{133}, 10_{137}, 10_{145}, 10_{161}, 10_{162}$
- 11 crossings: $11_{137}, 11_{139}, 11_{149}, 11_{156}, 11_{179}, 11_{190}, 11_{229}, 11_{231}, 11_{254}, 11_{257}, 11_{301}, 11_{317}, 11_{469}, 11_{471}, 11_{473}$

1.2 Conclusion and Conjecture

The calculation for 4_1 demonstrates how applying a knot group relator as an arithmetic operation sequence can lead directly to the vanishing condition for its Alexander polynomial. The computational results across numerous other knots significantly strengthen this observation, suggesting it is not an isolated coincidence but rather reflects a potentially deeper pattern.

This empirical evidence reinforces the conjecture that there is an underlying geometric or topological reason for this connection. It suggests that, for certain classes of knots or specific presentations, the arithmetic expression space framework H naturally encodes the constraint imposed by the Alexander polynomial. However, the precise conditions under which this correspondence holds, and its geometric interpretation within the structure of H (perhaps relating to the zero loci, torsion, or propagation mechanisms discussed previously), remain open questions requiring further investigation. Understanding the topological or algebraic properties that characterize the knots for which this direct correspondence holds is a key direction for future research exploring the intersection of arithmetic geometry and knot theory.